

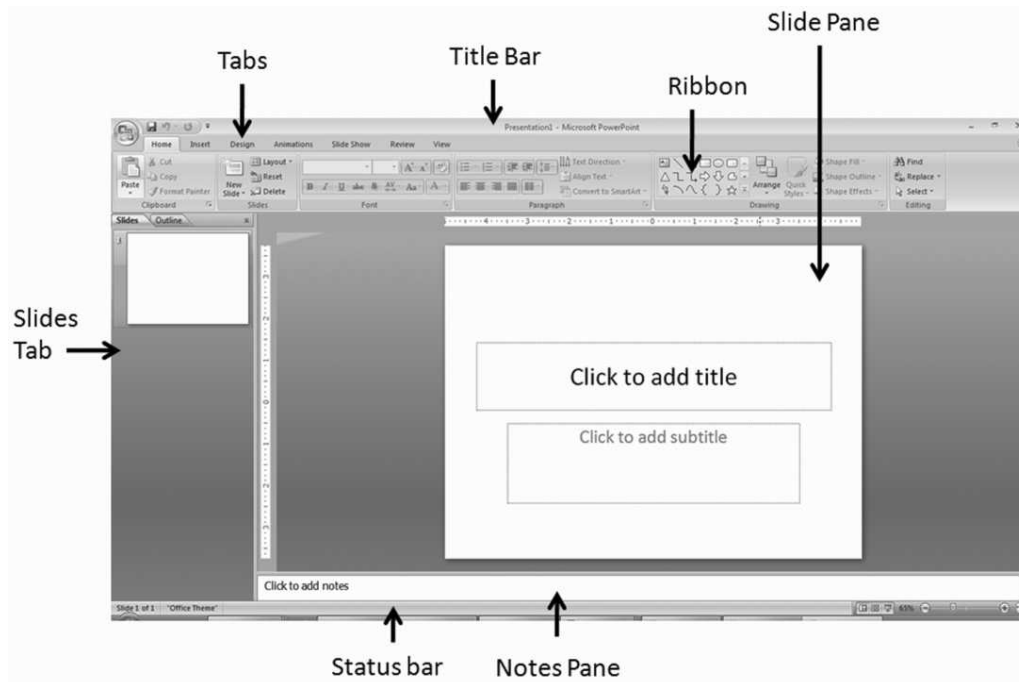


Fig. 1.21 The Microsoft PowerPoint—[Presentation1] Window

Using Run command We can also start MS PowerPoint by performing the following steps using the Run command:

1. Select Start → All Programs → Accessories → Run to display the Run dialog box.
2. Type powerpnt in the Open text box and click OK to display the Microsoft PowerPoint – [Presentation1] window.

1.13.2 Basic Operations Performed on a Presentation

The following are the key operations that can be performed in MS PowerPoint:

- ❖ Creating a new presentation
- ❖ Designing the presentation
- ❖ Saving a new presentation
- ❖ Adding slides to the presentation
- ❖ Printing the presentation

1.14 NETWORKING CONCEPTS

LO 1.7

Computer network is a system of interconnected computers that enable the computers to communicate with each other and share their resources, data and applications. The physical location of each computer is tailored to personal and organisational needs. A network may include only personal computers or a mix of

PCs, minis and mainframes spanning a particular geographical area. Computer networks that are commonly used today may be classified as follows:

- ❖ Based on geographical area:
 - Local Area Networks (LANs)
 - Wide Area Networks (WANs)
 - Metropolitan Area Networks (MANs)
 - International Network (Internet)
 - Intranet
- ❖ Based on how computer nodes are used:
 - Client Server Networks (CSNs)
 - Peer-to-peer Networks (PPNs)
 - Value-added Networks (VANs)

1.14.1 Local Area Network (LAN)

LAN is a group of computers, as shown in Fig. 1.22, that are connected in a small area such as building, home, etc. Through this type of network, users can easily communicate with each other by sending and receiving messages. LAN is generally used for connecting two or more personal computers through some medium such as twisted pair, coaxial cable, etc. Though the number of computers connected in a LAN is limited, the data is transferred at an extremely faster rate.

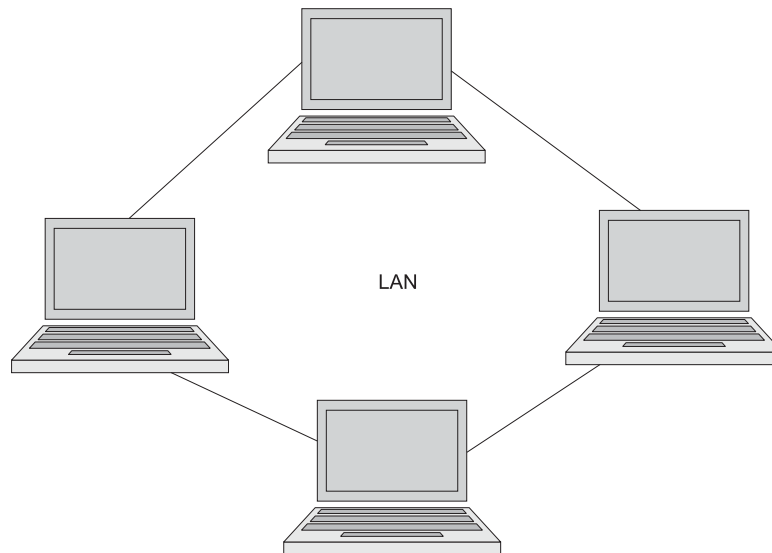


Fig. 1.22 A LAN

1.14.2 Wide Area Network (WAN)

WAN is a group of computers that are connected in a large area such as continent, country, etc. WAN is generally used for connecting two or more LANs through some medium such as leased telephone lines, microwaves, etc. In WAN, data is transferred at slow rate. A typical WAN network is shown in Fig. 1.23.

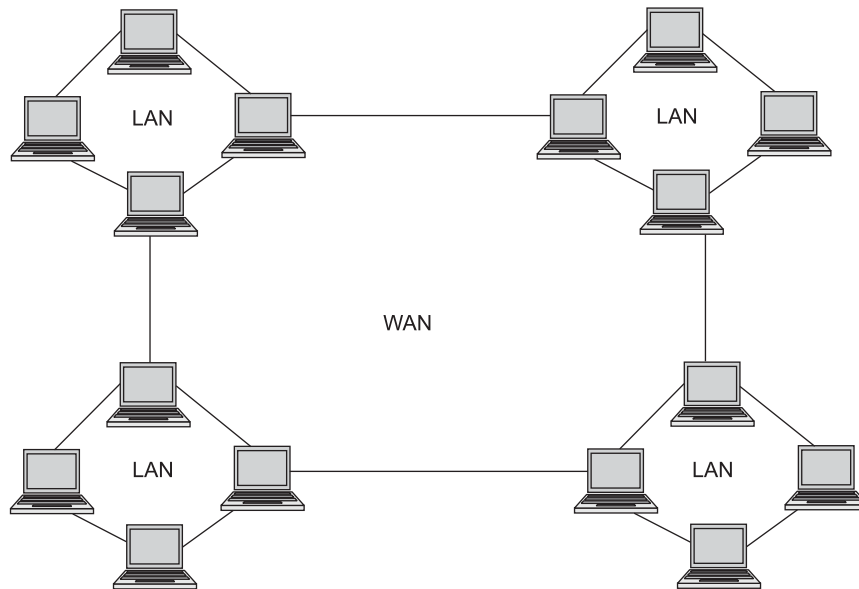


Fig. 1.23 A WAN system

1.14.3 Metropolitan Area Network (MAN)

MAN is a network of computers that covers a large area like a city. The size of the MAN generally lies between that of LAN and WAN, typically covering a distance of 5 km to 50 km. The geographical area covered by MAN is comparatively larger than LAN but smaller than WAN. MAN is generally owned by private organisations. MAN is generally connected with the help of optical fibres, copper wires etc. One of the most common example of MAN is cable television network within a city as shown in Fig. 1.24. A network device known as *router* is used to connect the LANs together. The router directs the information packets to the desired destination.

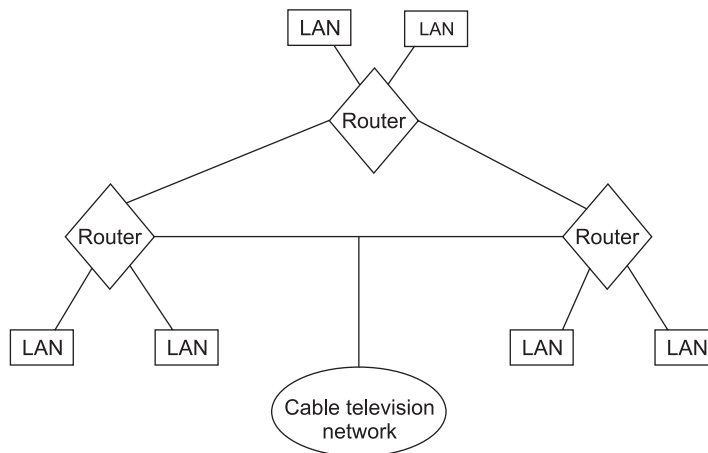


Fig. 1.24 A typical MAN system

1.15 NETWORK TOPOLOGIES

Network topology refers to the arrangement of computers connected in a network through some physical medium such as cable, optical fibre etc. Topology generally determines the shape of the network and the communication path between the various computers (nodes) of the network. The various types of network topologies are as follows:

- ❖ Hierarchical topology
- ❖ Bus topology
- ❖ Star topology
- ❖ Ring topology
- ❖ Mesh topology
- ❖ Hybrid topology

1.15.1 Hierarchical Topology

The hierarchical topology is also known as tree topology, which is divided into different levels connected with the help of twisted pair, coaxial cable or fibre optics. Figure 1.25 shows the arrangement of computers in hierarchical topology.

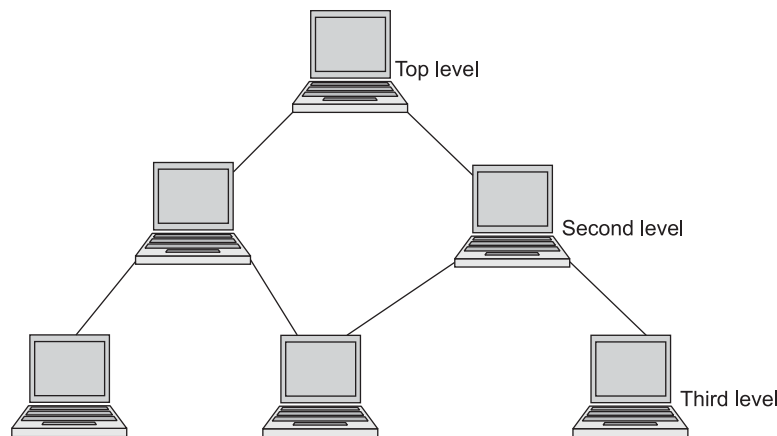


Fig. 1.25 *The hierarchical topology*

Advantages of hierarchical topology are:

- ❖ The hierarchical topology is generally supported by most hardware and software.
- ❖ In the hierarchical topology, data is received by all the nodes efficiently because of point-to-point link.

The following are the disadvantages of hierarchical topology:

- ❖ In the hierarchical topology, when the root node fails, the whole network crashes.
- ❖ The hierarchical topology is difficult to configure.

1.15.2 Linear Bus Topology

In the linear bus topology, all the nodes are connected to the single backbone or bus with some medium such as twisted pair, coaxial cable, etc. Figure 1.26 shows the arrangement of computers in the linear bus topology.

Advantages of linear bus topology are:

- ❖ The linear bus topology usually requires less cabling.
- ❖ The linear bus topology is relatively simple to configure and install.
- ❖ In the linear bus topology, the failure of one computer does not affect the other computers in the network.

The following are the disadvantages of linear bus topology:

- ❖ In the linear bus topology, the failure of the backbone cable results in the breakdown of entire network.
- ❖ Addition of computers in the linear bus topology results in the performance degradation of the network.
- ❖ The bus topology is difficult to reconstruct in case of faults.

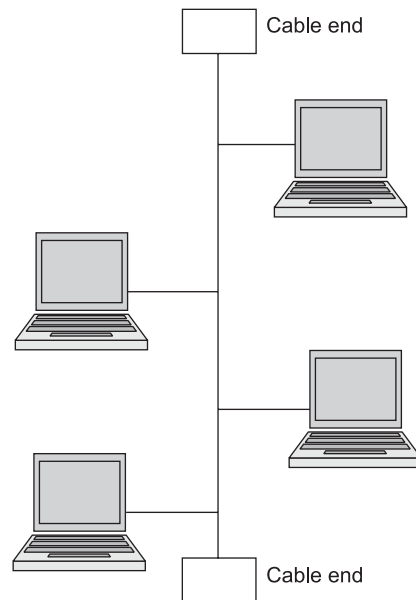


Fig. 1.26 A linear bus topology

1.15.3 Star Topology

In the star topology, all the nodes are connected to a common device known as hub. Nodes are connected with the help of twisted pair, coaxial cable or optical fibre. Figure 1.27 shows the arrangement of computers in star topology.

Advantages of star topology are:

- ❖ This topology allows easy error detection and correction.
- ❖ In the star topology, the failure of one computer does not affect the other computers in the network.
- ❖ Star topology is easy to install.

The following are the disadvantages of star topology:

- ❖ In the star topology, the hub failure leads to the overall network crash.
- ❖ The star topology requires more amount of cable for connecting the nodes.
- ❖ It is expensive due to the cost of hub.

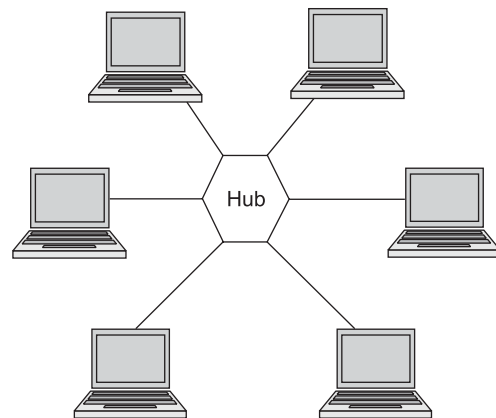


Fig. 1.27 A star topology

1.15.4 Ring Topology

In the ring topology, the nodes are connected in the form of a ring with the help of twisted pair. Each node is connected directly to the other two nodes in the network. Figure 1.28 shows the arrangement of computers in the ring topology.

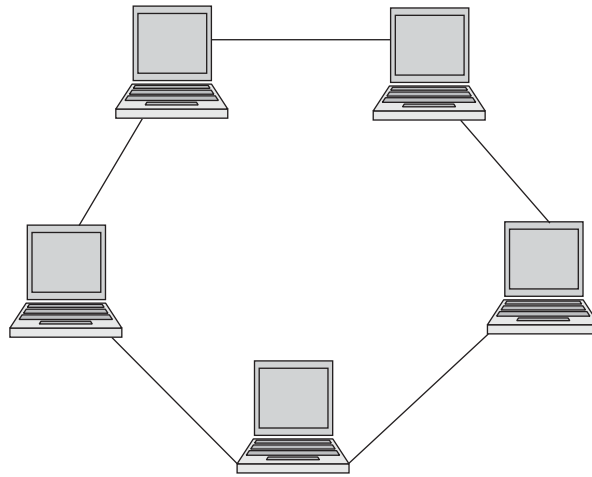


Fig. 1.28 A ring topology

Advantages of ring topology are:

- ❖ Each node has an equal access to other nodes in the network.
- ❖ Addition of new nodes does not degrade the performance of the network.
- ❖ Ring topology is easy to configure and install.

The following are the disadvantages of ring topology:

- ❖ It is relatively expensive to construct the ring topology.
- ❖ The failure of one node in the ring topology affects the other nodes in the ring.

1.15.5 Mesh Topology

In mesh topology, each computer is connected to every other computer in point-to-point mode as shown in Fig. 1.29. If we have n computers, we must have $n(n - 1)/2$ links.

Advantages of mesh topology are:

- ❖ Message delivery is more reliable.
- ❖ Network congestion is minimum due to large number of links.

The following are the disadvantages:

- ❖ It is very expensive to implement.
- ❖ It is very difficult to configure and install.

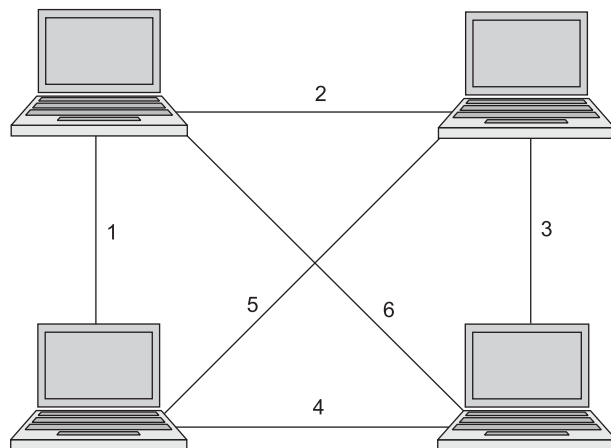


Fig. 1.29 Mesh topology

1.15.6 Hybrid Topology

The hybrid topology is the combination of multiple topologies, used for constructing a single large topology. Figure 1.30 shows a typical arrangement of computers in hybrid topology.

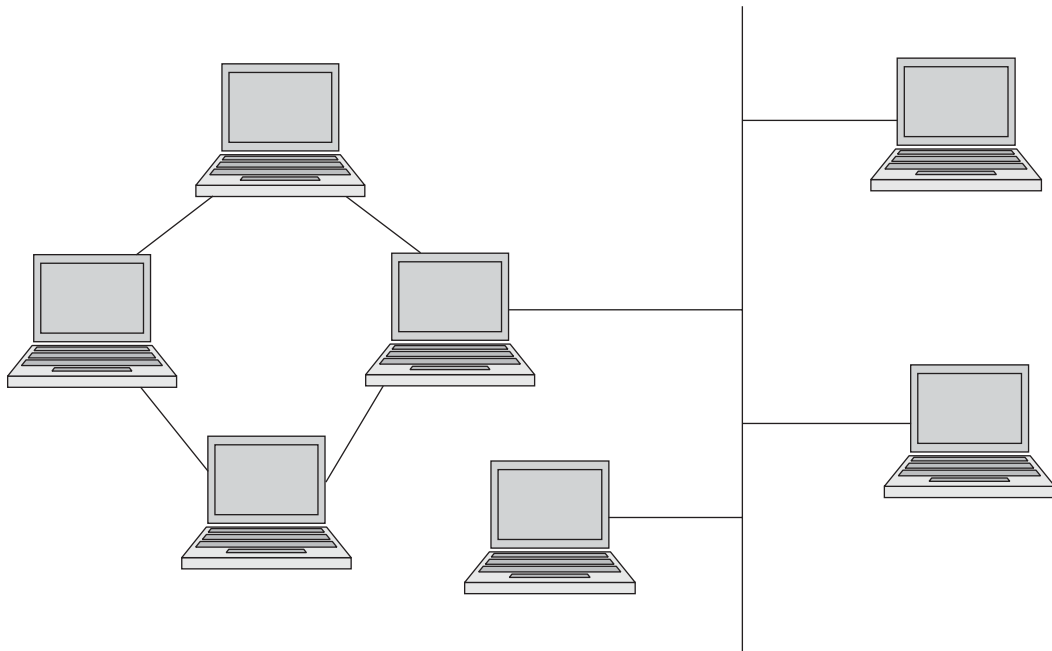


Fig. 1.30 A hybrid topology

Advantages of hybrid topology are:

- ❖ The hybrid topology is more effective as it uses multiple topologies.
- ❖ The hybrid topology contains the best and efficient features of the combined topologies from which it is constructed.

The following are the disadvantages of hybrid topology:

- ❖ The hybrid topology is relatively more complex than the other topologies.
- ❖ The hybrid topology is difficult to install and configure.

1.16 NETWORK PROTOCOLS AND SOFTWARE

LO 1.7

In order to share data between computers, it is essential to have appropriate network protocols and software. With the help of network protocol, computers can easily communicate with each other and can share data, resources, etc.

1.16.1 Network Protocol

Network protocols are the set of rules and regulations that are generally used for communication between two networks. Using network protocol, the following tasks can be performed:

- ❖ Identification of the type of the physical connection used
- ❖ Error detection and correction of the improper message
- ❖ Initiation and termination of the communication session
- ❖ Message formatting

Some of the commonly used network protocols are Hyper Text Transfer protocol (HTTP), Simple Mail Transfer Protocol (SMTP), File Transfer Protocol (FTP), Transmission Control Protocol/ Internet Protocol (TCP/IP), Telecommunications Network (Telnet), Domain Name System (DNS) etc.

HTTP

Hyper Text Transfer Protocol (HTTP) is the communication protocol used by the World Wide Web. It acts as a request-response protocol where the client browser and the Web server interact with each other through HTTP protocol rules. These rules define how messages are formatted and transmitted and what actions should the browser and Web server take in response to these messages. For example, when we type a URL in the address bar of a browser, then an HTTP request is sent to the Web server to fetch the requested Web page. The Web page details are transmitted to the client browser and rendered on the browser window through HTML.

In a typical situation, the client browser submits an HTTP request to the server and the server processes the request and returns an HTTP response to the client. The response contains status information pertaining to the request as well as the requested content (Figs 1.31–1.32).

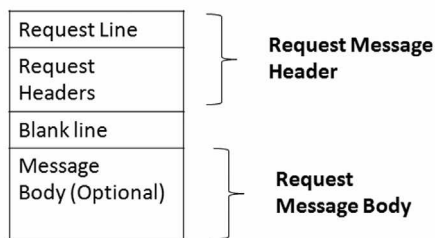


Fig. 1.31 HTTP Request Message format

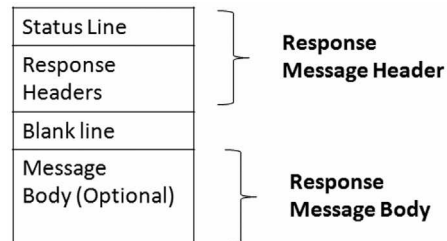


Fig. 1.32 HTTP Response Message format

HTTP protocol supports various methods that are used by the client browsers to send request messages to the server. Some of the common HTTP methods are:

- ❖ **GET**: Gets information from the specified resource
- ❖ **HEAD**: Gets only the HTTP headers
- ❖ **POST**: Posts information to the specified resource
- ❖ **DELETE**: Deletes the specified resource
- ❖ **OPTIONS**: Returns the list of HTTP methods that are supported by the Web server
- ❖ **TRACE**: Returns a diagnostic trace of the actions taken at the server end

The first line in an HTTP response object comprises a status line, which carries the response status code indicating the outcome of the HTTP request processed by the server. The status code is a 3-digit number and carries specific meaning, as described below:

- ❖ **1xx**: Comprises information status messages indicating that the server is still processing the request
- ❖ **2xx**: Comprises success status messages indicating that the request was received, accepted and processed by the server
- ❖ **3xx**: Comprises redirection status messages indicating that further action needs to be taken in order to process the request
- ❖ **4xx**: Comprises error status messages indicating error at client side, for example incorrect request syntax
- ❖ **5xx**: Comprises error status messages indicating error at server side, for example inability of the server to process the request

SMTP

Simple Mail Transfer Protocol (SMTP) is an e-mail protocol that is widely used for sending e-mail messages between mail servers. While SMTP supports capabilities for both sending and receiving e-mail messages, e-mail systems primarily used SMTP protocol for sending e-mail messages. For receiving, they use other protocols such as POP3 or IMAP. In Unix-based systems, sendmail is the most widely used SMTP server for e-mail. In Windows-based systems, Microsoft Exchange comes with an SMTP server and can be configured to include POP3 support.

FTP

File Transfer Protocol (FTP) is a standard protocol used for sharing files over the Internet. FTP is based on the client-server architecture and uses Internet's TCP/IP protocol for file transfer. The users need to authenticate themselves by specifying user name/password in order to establish a connection with the FTP server. However, some FTP sites also support anonymous login where users are not required to enter their credentials. To facilitate secure transfer of user's credentials and file contents over the Internet, FTP encrypts the content using cryptographic protocols such as TLS/SSL.

The following steps illustrate how file transfer happens through FTP:

1. The client machine uses Internet to connect to the FTP server's IP address.
2. User authentication happens by entering relevant user name and password.
3. Once the connection is established, the client machine sends FTP commands to access and transfer files. Now-a-days, various GUI-based FTP software are available that enable transfer of files through simple operations, such as drag and drop.

Telnet

Telnet is a protocol that allows users to connect to remote computers over a TCP/IP network, such as intranet or internet. While HTTP and FTP protocols are used for transferring Web pages and files over the Internet, the Telnet protocol is used for logging onto a remote computer and performing operations just as a normal user. The users need to enter their credentials before logging on the remote host machine.

Command-line based telnet access is available in major operating systems such as Windows, Mac OS, Unix and Linux. Generic format of the telnet command is given below:

Telnet host port

Here,

- ❖ **telnet:** Is the command that establishes telnet connection
- ❖ **host:** Is the address of the host machine
- ❖ **port:** Is the port number on which telnet services are available on the host machine



LEARNING OUTCOMES

- There are five generations of computer development which have seen tremendous shift in technology, size, and speed. **LO 1.1**
- On the basis of the size and capability, computers are categorized into microcomputers, mini computers, super computers and mainframe computers. **LO 1.2**
- Input devices help in inputting the data from any outside source into the computer system and output devices are used to pass on the processed data to the end users. **LO 1.3**
- Computer systems use two types of memory, namely primary memory and secondary memory. **LO 1.3**